

A Mini Project-1 Report on

SURGIASSIST

**Submitted to the Department of Computer Science & Engineering(AI & ML),
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B.Tech CSE (AI & ML) under JNTUH, Hyderabad**

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Certificate

This is to certify that the Mini Project-1 report on “**SURGIASSIST**” is a bonafide work carried out by **Peddisetti Ananya (23251A6619)**, **Ponugoti Tanvi Sri(23251A6620)**, **Shenishetty Bhavana(23251A6624)**, **Mallam Mahalaxmi(23251A6650)** in the partial fulfillment for the award of B. Tech degree in Computer Science & Engineering (AI & ML), G. Narayanamma Institute of Technology & Science, Shaikpet, Hyderabad, affiliated to Jawaharlal Nehru Technological University, Hyderabad under our guidance and supervision.

The results embodied in the Mini project work have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

SurgiAssist is a robust software solution that enhances the efficiency and management of hospital operations, focusing on both clinical and administrative functions. This project includes specialized features to support patients in the process of choosing and undergoing surgery. The system offers a detail of various surgeries, listing the types of surgeries available and the best hospitals equipped to perform those procedures. Most importantly, it integrates information about government healthcare schemes, enabling patients to find applicable financial support options for their surgeries.

SurgiAssist is a smart, easy-to-use software that helps patients and hospitals manage every step of the surgical process. For patients, it offers clear information about different types of surgeries, the best hospitals to go to, and the best schemes available. It even shows hospital location and detailed hospital profile so that patients can make informed choices they feel confident about.

Money is often a big concern when it comes to surgery. That is why SurgiAssist also connects patients with information about government healthcare schemes and financial aid options that they might qualify for — saving time, money, and a lot of stress.

Undergoing surgery can feel overwhelming — from choosing the right hospital and doctor to figuring out costs and financial aid. **SurgiAssist** is here to make that journey easier, safer, and a lot less stressful.

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1. INTRODUCTION

1.1 Motivation

Undergoing surgery is usually one of the scariest things a human being can go through — not merely physically, but emotionally, intellectually, and financially too. It is an experience of uncertainty, fear, and life-and-death choices, but most patients are left with next to no information about how to handle the ordeal. Numerous individuals feel suffocated by basic questions which should have very simple answers: Which hospital or physician can I trust with my life? How much is going to cost this procedure and if so, whether I am able to pay for it? Do I qualify for any government plans, insurance programs, or payment plans to assist me? These are not trivial fears — they're profoundly human anxieties that influence whether, when, or even if someone chooses to have life-saving or life-enhancing surgery. The healthcare system, too often, doesn't make it convenient to learn these things. Data is scattered across various channels, medical terminology can be confusing, and assistance seems out of reach or nonexistent.

That's what drove us to design SurgiAssist — an intuitive, intelligent, and effective tool designed with compassion and fueled by the conviction that each patient should have transparency, access, and respect in their healthcare experience. SurgiAssist is not merely a site; it's a patient's guide during one of the most vulnerable moments in their life. It collates key information all in one spot, allowing patients to find and compare hospitals in terms of quality, distance, and patient review. It refers them to veteran, reputable surgeons and specialists most appropriate for them. It expatiates procedure prices in ways easy to grasp, and — above all, probably — exposes payment plans they otherwise wouldn't know they can afford, like government plans, subsidization, and affordable-interest medical loans. It is all achieved by a neat, easy-to-use interface with the aim to alleviate stress rather than create more.

At its heart, SurgiAssist is about something more profound than logistics or numbers — it's about giving a patient a sense of control and confidence back. It's about taking fright and making sense of it. It's about turning confusion into understanding. It's about allowing patients to be heard, to be understood, and to be in control, to make informed decisions on their own behalf, and sometimes the behalf of their loved ones. For clinicians and hospitals, SurgiAssist is a route to improved patient participation,

streamlined administrative functions, and better care provision. By streamlining everything from scheduling appointments to paperwork to financial consulting, it allows medical centers to focus on what matters most — healing.

We created SurgiAssist because we believe healthcare must be human. We believe that by giving patients access to clear, timely, and empathetic guidance, outcomes are better, trust is established, and lives are changed. In a world where healthcare can too often be impersonal or even frightening, SurgiAssist is our way of saying: you are not alone, and there is a better path forward. Because when healthcare is clear, accessible, and human — everyone wins.

1.2 Existing Systems

Current systems offer at best fragmented and sometimes contradictory sources of information regarding surgery planning. Patients usually end up switching between hospital websites, scattered online reviews, Google Maps listings, and third-party sites — none of which provide an entire picture. Information on surgical techniques, hospital ratings, related fees, and insurance or financial assistance options is either fragmented, hidden in the small print, or not easily comparable from one institution to another. Such a disjointed environment requires patients to figure out everything themselves, an intimidating prospect under the duress and pressure of illness.

The absence of centralized, trusted information generates an intensely frustrating and disempowering experience. Patients usually spend hours — sometimes days — trying to determine which hospitals they can trust, which doctors do the surgery they require, and how much it will cost them. For most, particularly those from less advantaged backgrounds or with low levels of digital literacy, the process is not merely inconvenient but totally inaccessible. Essential information regarding low-cost treatment, government-run healthcare programs such as Ayushman Bharat or ECHS, and charitable or NGO-provided assistance is often overlooked or not pursued just because there is no single location to access it. Consequently, patients either forgo treatment, select inferior care due to cost issues, or receive surprise medical bills that lead them into financial distress.

The implications of this disjointed system are dire. Delayed decision-making can exacerbate medical conditions. Uncertainty about finances causes further emotional distress. Misconceptions regarding insurance coverage or support programs can result in patients not receiving the necessary care — or paying exponentially more than they ought to. The missing element is not merely a digital platform, but an integrated system prioritizing the patient. What there is, then, an acute need for, is an integrated platform which combines and makes accessible all information necessary to surgery — procedural, comparative hospital, doctor credentials, cost transparency, government schemes, and financing — in one coherent and easy-to-use experience.

Not only would this lessen patients' anxiety and confusion, it would enable them to make knowledgeable and timely decisions. It would be a bridge connecting healthcare providers, patients, and support systems with trust and enhanced outcomes. By bringing resources together and integrating healthcare planning in a more transparent, open manner, we can avoid unnecessary delays, curb surprise costs, and ultimately provide more individuals with better care they need, when they need it. Where digital technologies have transformed every aspect of life except healthcare, healthcare decision-making should be dealt with no less clarity, integration, and humanness.

1.3 Literature Survey

In the last few years, online health platforms have developed considerably, looking to increase hospital efficiency and patient care. While numerous tools aid administrative processes or clinical workflows, most of them are fragmented and do not have an integrated way of managing surgical care. Provider-centric hospital systems in the past provided little assistance to patients when it came to understanding surgeries or planning finances.

To fill these gaps, newer websites have added patient-centric features like treatment explanations and doctor directories. But they lack in providing end-to-end support—elements like hospital comparisons, location-based filtering, or access to government healthcare schemes. Financial constraints and unawareness of schemes like Ayushman Bharat still discourage patients from seeking timely surgical care.

SurgiAssist addresses these issues with a one-stop solution that integrates surgery catalogs, hospital profiling, geolocation-based search, and government aid scheme integration. Its intelligent, patient-focused design streamlines difficult choices, empowers patients with actionable data, and enhances the overall surgical experience through smart recommendations and system-wide transparency.

- In 2025, a study in the Journal of Medical Systems and at the IEEE Health Conference examined hospital operations and patient data integration. They employed system integration, web development, cloud technologies, and emphasized a great deal of UI/UX design. They encountered problems such as data privacy issues, intricate system integration, user adoption of the system, and high costs.
- In 2024, another paper in the International Journal of Healthcare IT presented a cloud-based system using APIs to merge into one system for hospitals. Although it held potential, it also had some challenges, predominantly with system fragmentation, possible security threats, and the requirement of further training of hospital personnel.
- In 2023, at the IEEE Health Informatics Conference, researchers presented a patient-centric healthcare management system. This system revolved around a patient portal and a backend support system for hospital functions. Though it had its benefits, it was challenged by integration issues, user experience, and high adoption costs.
- Last, a 2022 study published in the Journal of Medical Informatics centered on developing a thorough healthcare management platform by using machine learning for suggestions. Even though the concept of a centralized platform was novel, worries related to data integrity, regulatory compliance, and low rates of adoption proved to be huge hurdles.

1.4 Challenges to existing system

The modern healthcare system has patients competing for information along fragmented and uncoordinated pathways — hospital websites that deliver limited insight, Google Maps listings that deliver locations but little else, and surgery comparison sites that are shallow or worse, often inaccurate. None of these tools speak to each other, and none were designed with the patient's overall experience in mind. As a consequence, individuals making serious health choices are forced to cobble together

information from several sites, deciphering medical terminology, incomplete information, and conflicting guidance. This broken process is not only infuriating — it's hazardous. In moments when speed and clarity are most critical, patients are instead confronted with confusion, wasted hours, and growing anxiety.

Trying to compare surgery procedures, find hospital specialties, check doctor credentials, or estimate treatment costs is a chore so intimidating that too often it leads to doing nothing or making a bad choice. The emotional weight of a health issue is already heavy enough; adding uncertainty as to where to go to receive care, who to trust, and how to pay only serves to add to the weight. The absence of a centralized, simple website to use causes delay in treatment, as patients wait or reconsider. It causes unnecessary stress for patients and families already in weakened positions. More, financial uncertainty about medical procedures can lead to individuals not receiving necessary care at all, to the detriment of their health and well-being. Everyone is better than that. Everyone deserves a healthcare system that works with them, not against them — one that empowers, not overwhelms. Everyone needs an integrated, patient-centered platform where all the important information is laid out in plain sight, accurately and clearly, in one place. A tool that integrates surgery information, doctor and hospital comparisons, treatment schedules, estimated costs, insurance coverage, and access to government assistance, so patients can make informed decisions rapidly and with confidence. A platform that not only saves time, but earns trust — converting a confusing and fragmented process into one that feels guided, transparent, and human.

Without such a system, healthcare navigation doesn't feel quite like getting care, but rather getting lost in a labyrinth — one of dead ends, unexpected cost, and guessing. But with the right solution in place, we can redefine that experience entirely. We can turn uncertainty into clarity, loneliness into community, and fear into confidence. By doing so, we don't just improve individual patient experiences — we set a new standard for what healthcare must and should be in this new era.

1.5 Proposed System

SurgiAssist is an online healthcare platform that aims to streamline and consolidate access to essential surgical healthcare data in India. Amidst a disjointed and frequently opaque healthcare environment, SurgiAssist injects much-needed transparency by compiling surgery, hospital, and financial assistance program data into

a straightforward, integrated interface. Patients can browse categorized lists of surgeries and access hospital information with specifics on pricing, contact numbers, and maps to locations. Every hospital-surgery combination is displayed in a neat, standalone manner, enabling users to make quick and easy decisions. The responsive design offers easy access regardless of the device, whether an internet-aware customer or a newbie internet user.

Apart from static information, SurgiAssist also enhances users by embedding dynamic information like government healthcare plans like Ayushman Bharat and other state-managed insurance initiatives. These programs, usually under-promoted, have the potential to eliminate or cut significantly surgery fees for qualifying patients. SurgiAssist showcases these opportunities within hospital listings itself, with eligibility criteria and layperson descriptions. This makes the platform more than a search platform—it becomes a force for social equity, making necessary access to healthcare available for under-served groups. Secure login functionality allows users to set up individual accounts, bookmark favorite hospitals, monitor scheme applications, and receive reminders and updates, making the experience not only informative but interactive.

In the future, SurgiAssist is designed with scalability and long-term use in mind. The design of the platform is such that it allows for incorporating new features such as AI-based hospital suggestions, real-time appointment scheduling, insurance checks, multilingual interfaces, and feedback mechanisms. With the expansion of the platform, it will provide opportunities for hospital administrators and physicians, enable teleconsultation, and add rehabilitation referral. SurgiAssist is not a product—it's a compassionate health solution to cover knowledge gaps and bring in affordability, transparency, and accessibility during the planning of surgery across India. Through its sustained innovation and extension, it's going to be a strong constituent of the country's digital healthcare ecosystem.

1.6 Objectives

SurgiAssist aims to:

- provide users with an interface consisting of comprehensive list of surgeries and the hospitals where each surgery is available.

- display detailed cost information and contact details for hospitals offering specific surgeries.
- inform users about applicable health schemes available for each surgery at a particular hospital , hospital website and hospital location.
- integrate , test , and deploy the complete system.

1.7 Methodology

The creation of SurgiAssist was a systematic and user-oriented process, aiming to make surgical healthcare data readable and easily understandable to individuals from all walks of society. Right from the beginning, the aim was to integrate surgery, availability of hospitals, costs, and health scheme eligibility under one single, easy-to-use platform. A simple, responsive frontend was created with HTML, CSS, JavaScript, and Flask's Jinja2 template engine. It had an easy-to-use interface with a categorized list of surgical procedures and well-defined navigation paths, enabling users to easily transition from general surgery listings to detailed hospital and procedure pages. With responsiveness guaranteed across devices, the interface was made to be both visually pleasing and functionally efficient for varied user needs.

After the frontend was set up, backend integration started, driven by a MySQL database that was capable of dynamically loading and structuring data regarding hospitals, procedures, expenses, and government schemes. SQL JOIN queries enabled the fetching of related data in one request, providing users with immediate access to detailed procedure and hospital data. Flask managed the server-side functionality, translating user input into backend procedures and presenting the result through Jinja2 templates for uniformity and ease of scalability. With time, the system integrated functional features such as health scheme eligibility tags, web links to hospitals, and integrated Google Maps for easier navigation. The changes involved in the database schema and interface designs, but were gracefully implemented to make the system more usable without intimidating the user.

For system resilience, rigorous local testing occurred at all development phases. End-to-end tests, such as user authentication, data validation, and edge cases, guaranteed that the platform provided a consistent experience across features. Local

deployment allowed for rapid iterations and debugging, providing a solid base for the future migration to cloud platforms like AWS or Google Cloud. This migration will enable greater access, scalability, and cloud-native features such as load balancing and global reach. SurgiAssist evolved from a basic categorized frontend to a robust, integrated healthcare solution—constructed with care, tested with intensity, and engineered to create real-world impact by streamlining and democratizing access to surgical healthcare.

1.8 Hardware & Software requirements

Hardware: RAM , processor

Software : Python , JavaScript , MySQL , Django , Flask , HTML , CSS

1.9 Organization of the project:

The SurgiAssist project is divided into four layers: frontend, backend, database, and deployment. The frontend layer is developed with the use of HTML, CSS, JavaScript, and Flask templates (Jinja2), creating an amazing and responsive user interface where users can view the categorized list of surgeries and detailed hospital data such as costs, contact persons, health schemes, websites, and Google Maps locations. The backend, though composed, is developed in Flask (Python), and it serves the server-side logic, takes care of the navigation of users to service pages using defined routes such as /surgery_list and /surgery/, and interfaces the frontend with the database by executing SQL queries. The database layer is MySQL, where structured tables manage the storage of all the relevant information, and the SQL JOIN queries join surgeries data with hospital data, and with health scheme data, so as to display all the information joined dynamically on one page. Lastly, the deployment and testing layer brings all the layers of the project together. All the layers have been thoroughly tested to ensure each layer's functions work like login, signup, and manual retrieval all done as wished. An initial deployment for testing has been conducted locally with some future intentions to deploy on the cloud. All these layers have served as an easy platform for the display of surgery and hospital information with ease of access.

2.PROPOSED SYSTEM

2.1 Architecture

The architecture of the proposed model is shown in the figure 1.0 below

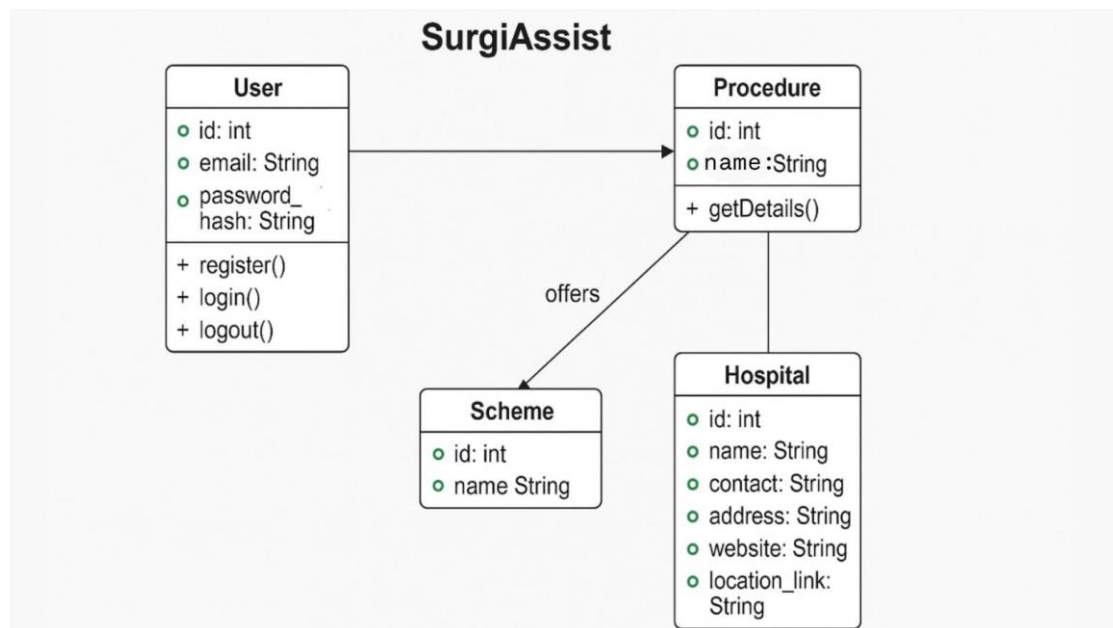


Figure 2.1.1 UML diagram of the model

The SurgiAssist system is designed to help users explore surgical procedures, the hospitals that offer them, and the healthcare schemes available. The system consists of four main entities: **User**, **Procedure**, **Hospital**, and **Scheme**. A User can register, log in, and log out of the system using secure credentials. Procedures represent surgeries or medical treatments and can be explored through a method that retrieves detailed information. These procedures are offered by various Hospitals, each of which contains details like name, contact information, address, website, and a location link. Some hospitals may also offer specific procedures under government or private healthcare Schemes, which are represented by their names and unique identifiers. The relationships between these entities enable users to find which hospitals perform a particular procedure and whether any scheme coverage is available, supporting informed medical decisions.

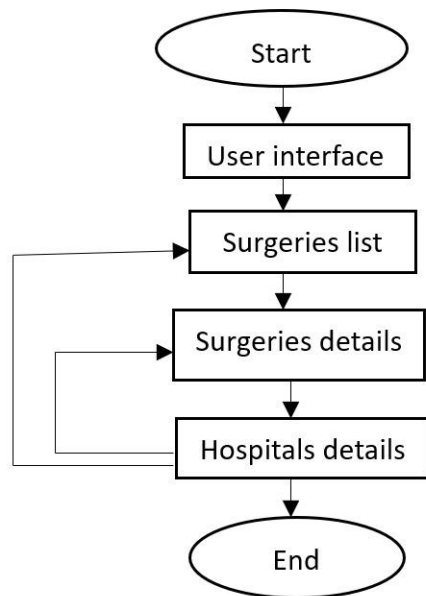


Figure 2.1.2 Architecture Flow of the model

2.2 Module level design

1. User Management

- Sign-up Page
- Login Page

2. Surgery and Hospital Directory

- List and search surgical procedures
- View hospitals linked to each procedure

3. Hospital Details and Cost Viewer

- Show hospital name, contact number, address, and cost of procedure
- Include links to hospital websites
- Embedded map/location link support

4. Scheme Coverage and Benefit Information

- Show schemes applicable per hospital-procedure pair
- Filter hospitals based on scheme availability

5. System Integration and Testing

- Integrate frontend, backend, and database
- Perform unit and integration testing

3. IMPLEMENTATION AND TESTING

3.1 Implementation

Project Setup

We initiated the SurgiAssist platform using Flask, a lightweight web framework in Python. A virtual environment was created using `python -m venv venv`, activated via `source venv/bin/activate`, and the required packages were installed using `pip install -r requirements.txt`. The dependencies included flask and mysql-connector-python.

We connected our Flask application to a MySQL database named `surgi1`, using credentials configured in `app.py`. A secure `secret_key` was set for session management.

User Authentication

User sign-up and login were handled securely:

- Passwords were hashed using Werkzeug's `generate_password_hash` before storing in the `users1` table.
- Authentication was performed using `check_password_hash`.
- Sessions were used to manage user login state.

Signup and login templates (`signup.html` and `login.html`) were created with basic form validation and error handling using Jinja templating

Views and Routes

The application's core functionality was implemented in `app.py` using Flask routes:

- `/signup`, `/login`, and `/logout` handled authentication.
- `/surgery_list` displayed a searchable list of surgeries.
- `/surgery/<name>` showed detailed information about hospitals offering the selected surgery.

Each route was protected to ensure only authenticated users could access sensitive pages.

Templates

HTML templates using Jinja2 were stored in the templates/ directory:

- surgery_list.html presented a searchable table of procedures.
- surgery_details.html displayed hospitals filtered by cost and insurance scheme.
- hospital_details.html showed additional cost and scheme-related data for selected hospitals.

These pages used embedded CSS and minimal JavaScript for interactivity.

Filtering and Dynamic Rendering

We added filtering options in surgery_details.html for:

- Minimum/Maximum Cost
- Scheme Coverage (covered or not)

Flask dynamically processed these inputs on the backend and returned filtered data from joined MySQL tables like procedures, procedure_offerings, hospitals, schemes, and scheme_coverage

Database Integration

SurgiAssist is tightly integrated with a MySQL database that serves as the central repository for all application data. The system follows a relational model, consisting of multiple interrelated tables. These include tables for surgeries, hospitals, procedure offerings, insurance schemes, coverage information, and user credentials.

The structure is designed to be normalized, reducing data redundancy and ensuring consistency. Relationships between entities such as surgeries and hospitals are managed using foreign keys and join tables. For example, the linkage between procedures and hospitals is handled through a mapping table that also stores cost information, while scheme coverage is represented through a separate table that connects procedures to government or private schemes.

The application logic is written to handle real-time queries that retrieve and process data based on user inputs, such as cost range filters or scheme availability. Efficient

SQL queries are used to pull data by combining multiple tables and applying conditional logic to generate tailored views for users.

Static Files

In SurgiAssist, static files play a crucial role in enhancing the appearance and interactivity of the web interface. These files include CSS for styling, JavaScript for client-side functionality, and any necessary image or media assets. All static assets are stored in a dedicated static directory and referenced from within HTML templates.

The CSS files are responsible for defining the visual layout, such as the structure and responsiveness of the login and signup forms, tables listing surgeries, and detail views of hospitals. They ensure the application is clean, accessible, and visually appealing.

JavaScript is used to introduce client-side behaviors. For instance, users can dynamically search for surgeries from the list without needing to refresh the page. Additionally, it allows toggling between different sections of content, such as expanding or collapsing detailed hospital views based on user interaction.

Security

We prioritized security by:

- Hashing passwords before storage.
- Using sessions to protect routes.
- Validating form input on both frontend and backend.

Deployment

Although not fully deployed, the application is structured to be ready for deployment on platforms like Heroku, Render, or AWS EC2. This includes:

- Environment-specific configurations (e.g., `secret_key`, DB credentials).
- Static file serving using Flask's `send_from_directory()` if needed in production.

3.2 Testing

Unit Testing: Individual functions like login, signup, and database queries were manually tested during development.

Integration Testing: Ensured that frontend forms, routes, and database operations work seamlessly.

Functional Testing: Confirmed that major functionalities like cost filtering, scheme coverage display, and hospital linking operate correctly.

Security Testing:

Passwords are hashed before storage.

Sessions prevent unauthorized access to protected pages.

Manual UI Testing

Tested navigation from:

Signup → Login → Surgery List → Surgery Details → Hospital View

Checked edge cases such as:

Login with wrong password

Searching for surgeries that don't exist

Filtering with invalid or missing inputs

3.3 Description of Technologies/Tools used

Frontend

HTML & CSS: Used for structuring and styling all user-facing pages including signup, login, surgery list, and hospital details.

JavaScript: Enables interactivity such as live search filtering in the surgery list and toggling hospital detail views.

Jinja2 (Flask Templating Engine): Allows embedding Python logic into HTML for dynamic content rendering.

Backend

Python: Primary programming language used in conjunction with Flask for server-side logic and route handling.

Flask: Lightweight Python web framework used to manage routes, handle requests/responses, and connect with the database.

Database

MySQL: Used as the relational database to store and manage data related to users, hospitals, surgeries, procedures, and schemes.

Authentication

Werkzeug Security: Provides secure password hashing and verification using `generate_password_hash()` and `check_password_hash()`.

Other Tools

MySQL Connector for Python: Facilitates database interaction within the Flask application.

Git/GitHub: For version control and collaborative development.

Local Development Server: Used for testing and debugging using Flask's built-in server.

3.4 Modules Description

1. User Management

Sign-Up: Allows new users to register by securely storing their hashed credentials.

Login: Authenticates returning users and creates a session.

Session Management: Ensures pages like surgery listing and details are only accessible to logged-in users.

2. Surgery & Hospital Directory

Surgery List: Shows a searchable table of surgeries offered.

Details Page: Displays hospitals that offer the selected surgery, with pricing, scheme coverage, and contact information.

3. Hospital Details Viewer

Extended View: Offers detailed information like address, website, and Google Maps location in a pop-up view.

Navigation: Users can toggle between the surgery list and hospital detail views.

4. Scheme Integration Module

Scheme Display: Shows which healthcare schemes (like Ayushman Bharat) are applicable to surgeries at specific hospitals.

Filtering: Lets users filter hospitals by scheme coverage and cost range.

5. Database Management

Tables: Users, procedures, hospitals, offerings, schemes, and scheme coverage.

Relations: JOIN queries link data across multiple tables for real-time dynamic rendering.

6. Static File Handling

CSS: Defines visual appearance.

JavaScript: Adds interactivity and improves user experience.

4. Results and Description

The development and implementation of the *SurgiAssist* platform yielded valuable results, demonstrating its potential to assist patients and caregivers in accessing critical surgery-related information. The application's ability to centralize, filter, and present healthcare data such as surgery costs, hospital availability, and scheme coverage has made healthcare navigation more transparent and user-friendly.

One of the key outcomes of the platform is the efficient aggregation of surgery-related information. Users can now explore a comprehensive list of surgeries, view associated hospitals, and filter based on cost and government scheme coverage. This significantly reduces the time and effort required to manually search for such details across different sources.

User interactions within the system indicate a strong reliance on cost filtering and scheme-based sorting, suggesting that affordability and financial assistance are crucial decision-making factors for most users. The backend query system, designed to handle these filters dynamically, proved effective in returning relevant and concise data to the end user.

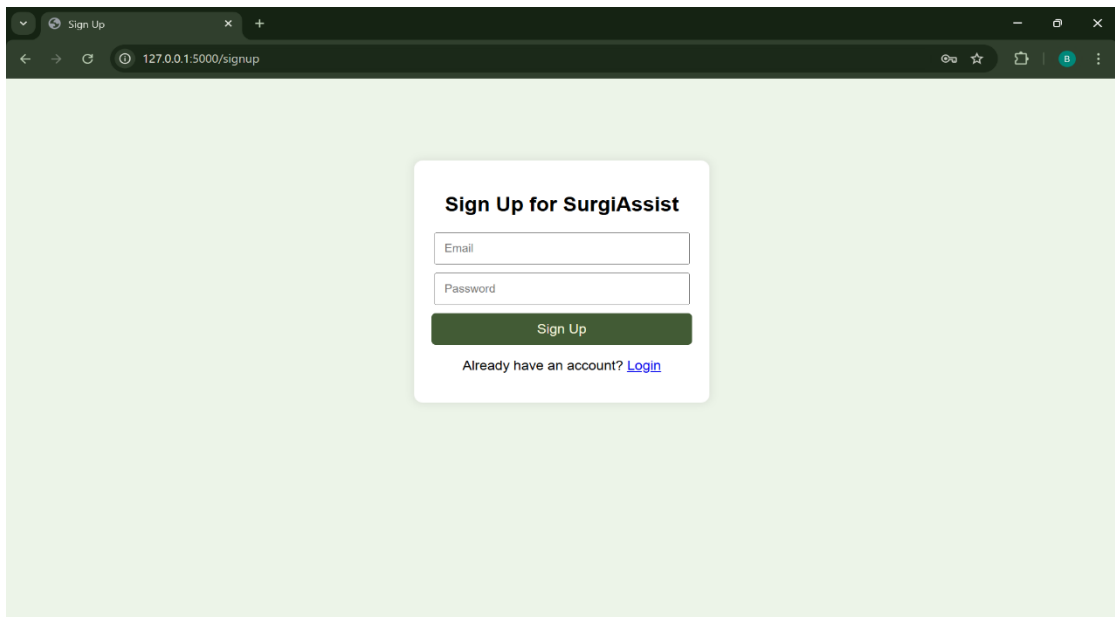
From these results, several inferences can be drawn:

- The presence of filter tools enables patients to make informed decisions based on financial constraints.
- Hospitals with scheme coverage and reduced cost options tend to be more frequently accessed, indicating a user preference for subsidized or scheme-affiliated institutions.
- The simple login and signup system supports secure and personalized access, which increases user trust and encourages system usage.

These insights underline the need for ongoing data enrichment, including real-time hospital availability and integration of more schemes. Future updates may include user feedback collection to further personalize search results and improve platform usability.

4.1.1 Sign Up Page

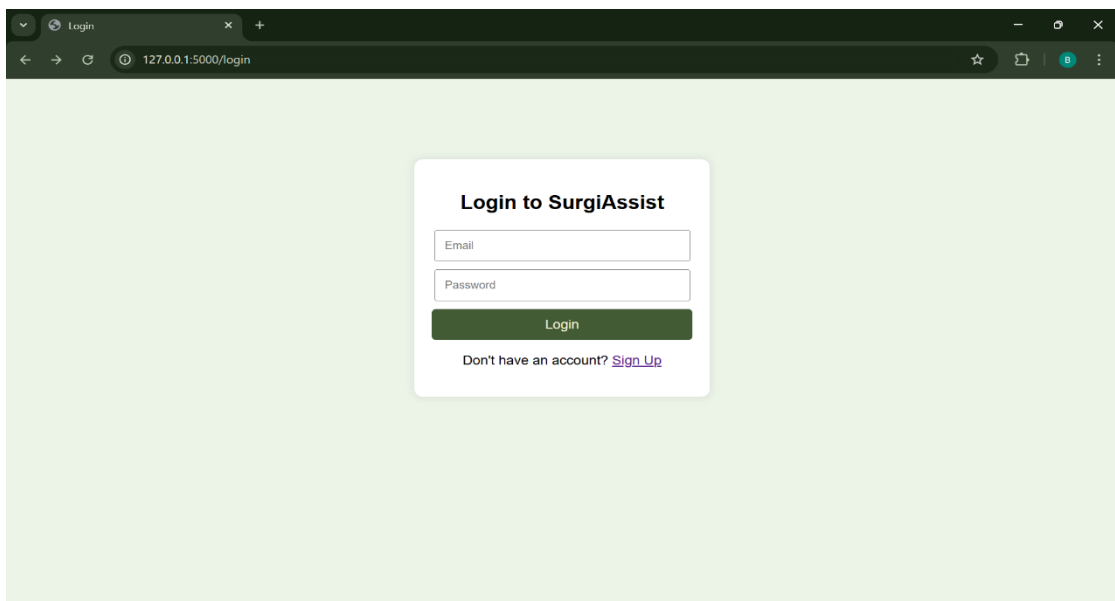
New users can create an account by providing their email and password. This ensures secure access to surgery-related information and enables personalized user sessions.



(Figure 4.1.1 Sign Up Page)

4.1.2 Login Page

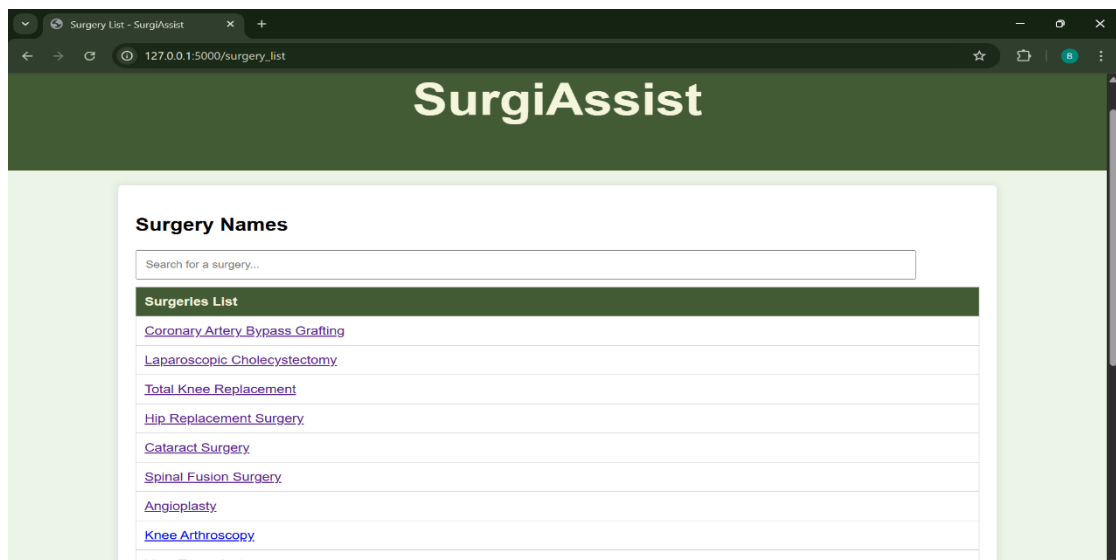
Returning users can log into the platform by entering their credentials. Error handling ensures that invalid login attempts are clearly communicated to the user.



(Figure 4.1.2 Login Page)

4.1.3 Surgery List Page

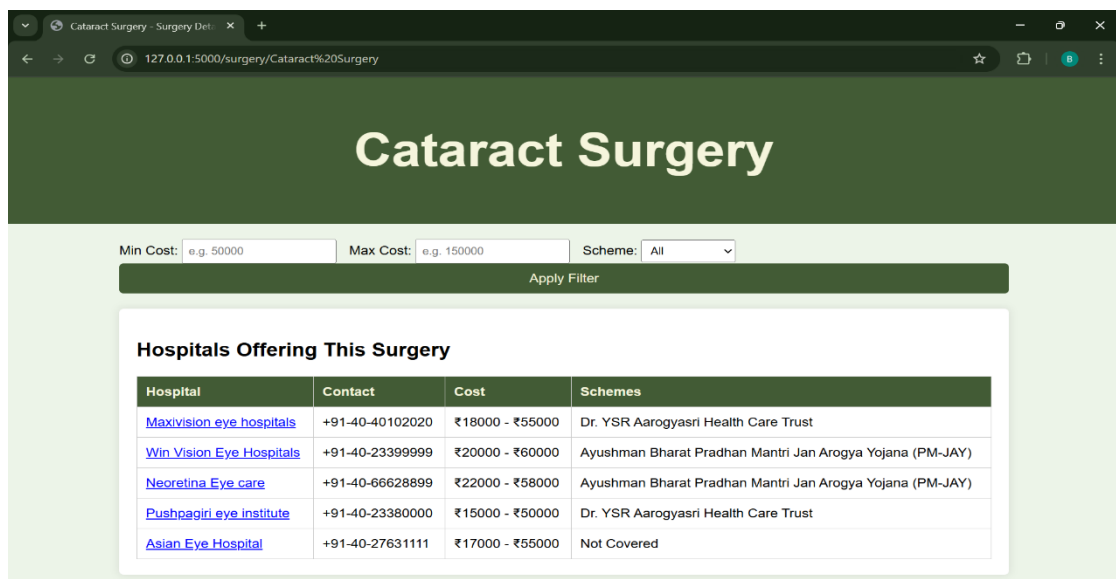
Once logged in, users are directed to a searchable list of surgeries. They can browse or filter surgeries by name, improving navigation and accessibility.



(Figure 4.1.3 Surgery List Page)

4.1.4 Surgery Details Page

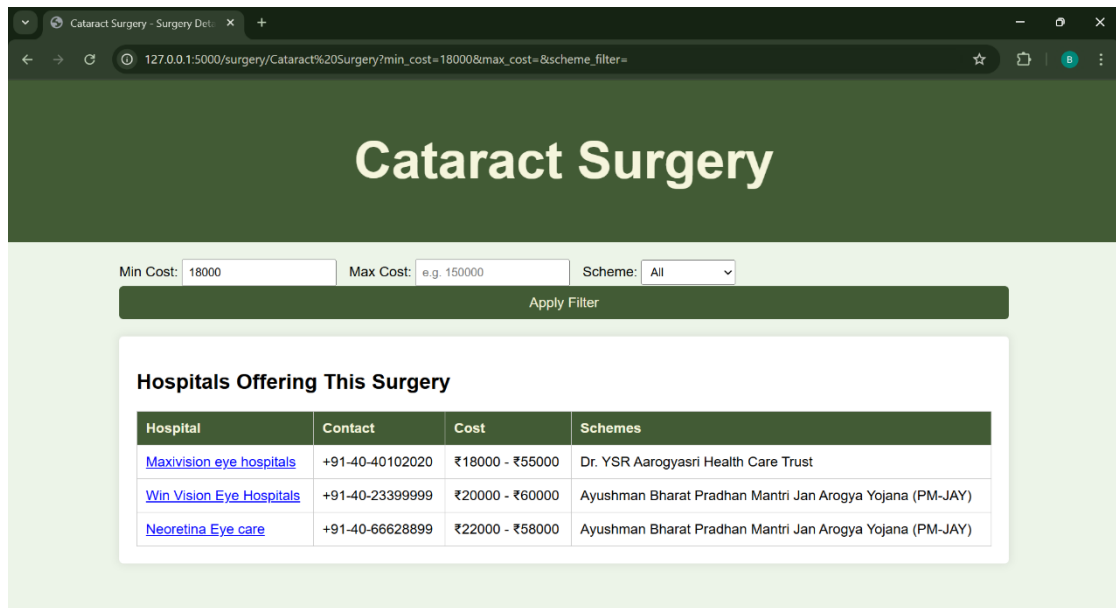
Selecting a surgery shows a list of hospitals offering the procedure, including contact info, cost ranges, and scheme availability. Filters for cost and scheme type help refine the results.



(Figure 4.1.4 Surgery Details Page)

4.1.5 Filter Functionality Panel

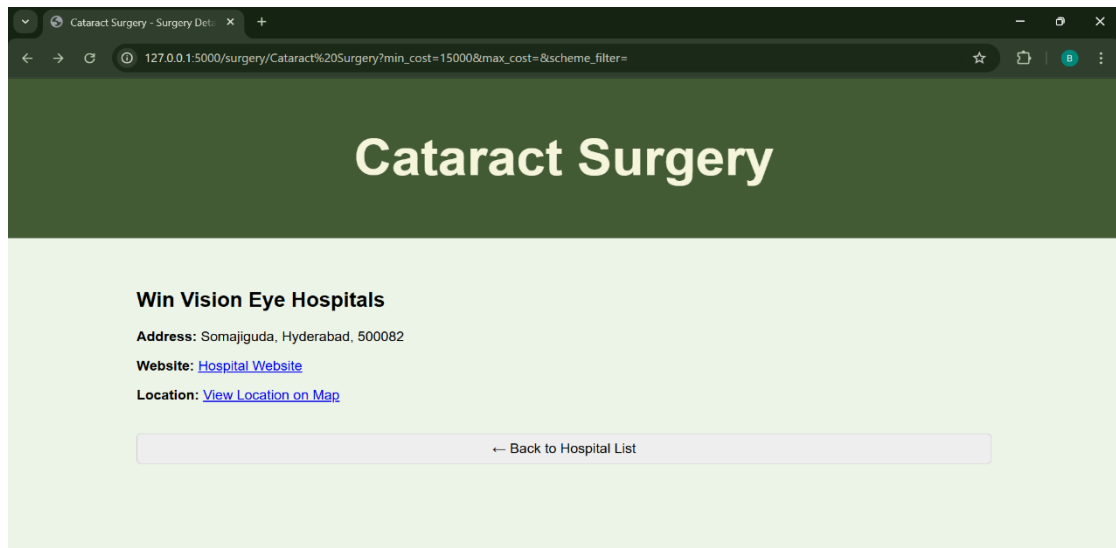
This panel allows users to refine hospital search results based on minimum and maximum cost and scheme coverage options. It enhances decision-making by displaying only relevant, user-specific data.



(Figure 4.1.5 Filter Functionality Panel)

4.1.6 Hospital Detail View

Clicking on a hospital reveals more details like address, website, and map location. This view helps users gather logistical information for hospital visits or consultations.



(Figure 4.1.6 Hospital Detail View)

5. CONCLUSIONS AND FUTURE WORK

5.1 Conclusion

SurgiAssist is a holistic and patient-focused digital platform that aims to revolutionize the manner in which people access, comprehend, and navigate surgical healthcare services. Through the consolidation of enormous amounts of critical information into a single consolidated interface, SurgiAssist solves the perennial problem of disconnected healthcare information, which has a natural propensity to confuse, overwhelm, or mislead patients.. The website integrates an exhaustive and searchable database of surgeries, an elaborately researched list of hospitals and clinics where each surgery is carried out, unbiased cost comparison from the hospitals, and essential information about the hospital such as contact numbers, physical addresses, websites, and Google Maps location for accessibility. It further provides the related health schemes, government schemes, and insurance schemes to facilitate users in making economically rational decisions.

With inclusiveness and accessibility built into its design, SurgiAssist has simple search and filtering features so that users can search for surgeries by physician specialty, location, cost, and patient rating. It also has educational content that explains procedures, encapsulates pre-operative and post-operative procedures, and includes recovery timetables, so patients are better informed and more confident about their choices. For the user who happens to be located in rural or under-served regions, offline capabilities (coming soon) and multilingual access offer the choice available for use whether or not the person has access to the internet or proper digital literacy. SurgiAssist not just empowers the patient but also the physician via automated patient intake and acquisition by appointment scheduling integration and verifications that are documented.

Lastly, SurgiAssist closes the gap between information and action by allowing patients to make timely, informed, and transparent health care decisions. By consolidating all of the information that the patient should be aware of regarding surgery in one convenient place, the platform enhances the health care experience as a whole, eliminates unnecessary delay and cost, and allows for improved outcomes for health.

5.2 Future Work

SurgiAssist is set to revolutionize surgical healthcare with a holistic digital platform that simplifies the whole patient experience—from diagnosis to recovery. In the near

future, it will provide real-time consultation and surgery booking, solving the age-old problem of delays and inefficiency in scheduling healthcare. Patients can look for and book appointment slots with specialists, hospitals, diagnostic laboratories, and recovery units from the comfort of their own homes. Extensive integration with large insurance companies will enable patients to verify coverage, see co-payment information, and get true cost estimates in real-time. Decision-making will be supported by AI-driven real-time suggestions taking into account medical requirements, cost, location, and patient opinions to suggest the most appropriate procedures and hospitals.

The platform will extend beyond the surgery itself, with continuity of care through post-surgery care such as rehabilitation, home nursing, physiotherapy, follow-ups, and telemonitoring. The integrated approach will provide patients with advice and reminders, and complications will be dealt with promptly. Foreign and rural patients will enjoy multilingual telemedicine services, virtual consultations, and treatment coordination that minimizes travel and cost. With dynamic cost monitoring, users will be constantly exposed to current prices, offers, and promotions, for complete transparency. In the meantime, the secure health record management system will enable patients to easily share medical information, with upcoming integration of blockchain to further ensure data integrity and traceability.

SurgiAssist also focuses on education and equity. A specialized learning center will provide content like surgical tutorials, tips for recovery, and Q&A with physicians to mentally and physically prepare patients. In collaboration with government health initiatives like Ayushman Bharat, the platform will strive to make affordable surgical care available to marginalized populations. Voice navigation, low-bandwidth optimization, and multi-language support features will make the platform accessible to senior citizens and digitally marginalized individuals. Finally, SurgiAssist guarantees not only a booking facility, but a comprehensively integrated, patient-focused healthcare experience—revolutionizing surgical healthcare into an educated, transparent, and empathetic journey for all users.

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GLOSSARY

SurgiAssist:

A digital healthcare platform designed to centralize and simplify access to surgical procedure information, hospital availability, pricing, and government healthcare schemes in India.

Surgery List Page:

A user interface that displays a comprehensive list of surgical procedures, allowing users to search or browse surgeries and view associated hospital details.

Surgery Details Page:

A page that shows a filtered list of hospitals offering the selected surgery, along with costs, contact details, scheme applicability, and map links.

Hospital Directory:

A catalog within the platform that includes hospital names, locations, contact information, websites, and procedures offered.

Health Schemes:

Government or private financial aid programs like Ayushman Bharat or ECHS that subsidize surgery costs for eligible patients.

Procedures Table:

A database table that stores information about various surgical procedures.

Hospitals Table:

A MySQL table storing hospital data such as name, contact, address, website, and location.

Procedure Offerings:

A mapping table linking surgeries to the hospitals that provide them, including fields like minimum and maximum cost.

Scheme Coverage:

A table that represents the relationship between hospitals' surgical offerings and the healthcare schemes covering them.

Flask:

A lightweight Python web framework used to build the backend of SurgiAssist, handle routing, and integrate with the database.

Jinja2:

The templating engine used with Flask to dynamically generate HTML pages with embedded Python logic.

MySQL:

A relational database management system used to store structured healthcare data in SurgiAssist.

User Authentication:

A security mechanism involving user sign-up, login, and session management to restrict access to certain features.

Session Management:

Used in Flask to maintain a logged-in user's state across pages during a browsing session.

SQL JOIN Queries:

Database operations used to retrieve related data from multiple tables in a single request, enabling dynamic rendering of complex information.

Cost Filter:

A user input mechanism allowing filtering of hospital listings based on minimum and maximum surgery costs.

Static Files:

CSS, JavaScript, and image files stored separately to handle the front-end appearance and interactivity of the platform.

Responsive Design:

Design principle ensuring that the platform adjusts gracefully to different screen sizes, improving usability on mobile and desktop devices.

Deployment:

The process of making the web application available to users on the internet, possibly using platforms like Heroku or AWS.

APPENDIX

Abbreviations:

HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
SQL	Structured Query Language
API	Application Programming Interface
UI/UX	User Interface/User Experience
IEEE	Institute of Electrical and Electronics Engineers
NGO	Non-Governmental Organizations